Two-minute cadence for a pre-registered transit of the 36.3-day sub-Neptune candidate TIC 402710280.01

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1 DDT Justification

TIC 402710280 (TYC 9041-03677-1; $T = 9.24$, $V = 9.98$) hosts TIC 402710280.01, a transiting sub-Neptune candidate with $P = 36.33986 \pm 0.00051$ d (Vora, 2026a). *TESS* has observed the star only in full-frame images (FFIs). The period was fixed only by the consecutive transits of Sectors 100 and 102 (2026 February and March), so the target was not proposed in the Cycle 9 Guest Investigator call. *TESS* will observe the star in Sectors 113 and 114 and, according to `tess-point` (Burke et al., 2020), not again through Sector 134 (2028 September). The transit of 2027 February 19, in Sector 114, is therefore the only one *TESS* will record before late 2028. Without a DDT allocation it will be recorded in 200-s FFIs only, with no target pixel files and no SPOC light curve.

2 Scientific Justification

Four transits (Sectors 39, 93, 100 and 102; Figure 1), measured in three independent reductions (TESS-SPOC, QLP and our own FFI photometry), share a depth of 2092 ± 47 ppm and a duration of 5.27 ± 0.05 h. Every shorter alias of the period predicts transits in covered data that are absent at $\geq 18.6\sigma$. For the TIC stellar parameters ($T_{\rm eff} = 5182$ K, $R_\star = 0.82\,R_\odot$, $M_\star = 0.88\,M_\odot$), the candidate has $R_p = 3.75 \pm 0.22\,R_\oplus$, $a = 0.206$ AU and $T_{\rm eq} \approx 500$ K. Difference-image centroids place the signal $1.5'' \pm 1.3''$ from the target. The only *Gaia* source near enough to contribute, $3.5''$ away and 354 times fainter, would need a 74% eclipse, whose shape is disfavoured for any main-sequence pair ($\Delta\chi^2 \geq 142$). No secondary eclipse is detected. `TRICERATOPS` (Giacalone et al., 2021) gives FPP $= 0.003 \pm 0.004$ and NFPP $< 10^{-6}$; without high-resolution imaging we treat these as supporting evidence, not as validation. At $V = 10$ the host suits radial-velocity follow-up: a mass of about 14 $M_\oplus$ would induce a semi-amplitude of about 2.8 m s^{-1}.

Two-minute data in Sectors 113 and 114 would serve three purposes.

(1) The pre-registered test. The transit times scatter about a linear ephemeris by 21.8 min rms, against formal uncertainties of 1.7–3.3 min. We find no instrumental cause, and four epochs cannot separate timing systematics from transit-timing variations. On 2026 October 15 we registered the epoch-57 prediction ($\mathrm{BJD_{TDB}}$ 2461455.978 $\pm$ 30 min), the data to be used, and the criteria that classify the outcome (Vora, 2026b). The protocol names SPOC two-minute light curves as its first-choice data, and its check of the signal's position requires target pixel files.

(2) An independent mid-time. The largest contribution to the scatter is the interval between the Sector 100 and 102 transits, both from our own photometry, which exceeds P by 29 min. A SPOC mid-time at epoch 57, independent of our reductions, tests whether that interval reflects our photometry or the orbit.

(3) The 3.78-day signal. The light curve also contains a 407 ± 26 ppm, 1.3-h signal with $P = 3.78371$ d, whose source, the target (as a near-grazing transit) or the $3.5''$ neighbour, the present data cannot identify (Vora, 2026a). The two sectors contain about a dozen of its events. Two-minute cadence would resolve their ingress, which tests the grazing geometry, and the target pixel files would tighten the difference-image centroid, now $5.2'' \pm 6.7''$ from the target.

3 Target Justification and Feasibility

We request two-minute cadence for one target, TIC 402710280 (RA 248.5236°, Dec −63.5497°), in Sectors 113 and 114, at priority 1.0. `tess-point` places it on camera 4, CCD 4 (column 385, row 209) in Sector 113 and on camera 4, CCD 1 (column 646, row 475) in Sector 114. The TIC contamination ratio is 0.06, and the target is fainter than 7th magnitude, so it needs no enlarged pixel stamp.

The epoch-57 transit is predicted for 2027 February 19 at 11:29 UTC with $\sigma = 30$ min (34 min when the covariance of T_0 and P is included), six days after the start of Sector 114 and well inside its first orbit. The neighbouring transits, on 2027 January 14 and March 27, fall outside both sectors.

In FFIs each transit depth is measured to 86–144 ppm, so a single transit is detected at 15–24σ; SPOC two-minute photometry should match or exceed this. Ingress and egress last about 14 min each, which is seven two-minute cadences against four 200-s cadences. The finer sampling sharpens the mid-time and the ingress shape, which is what distinguishes a planet from a diluted eclipse of the neighbour.

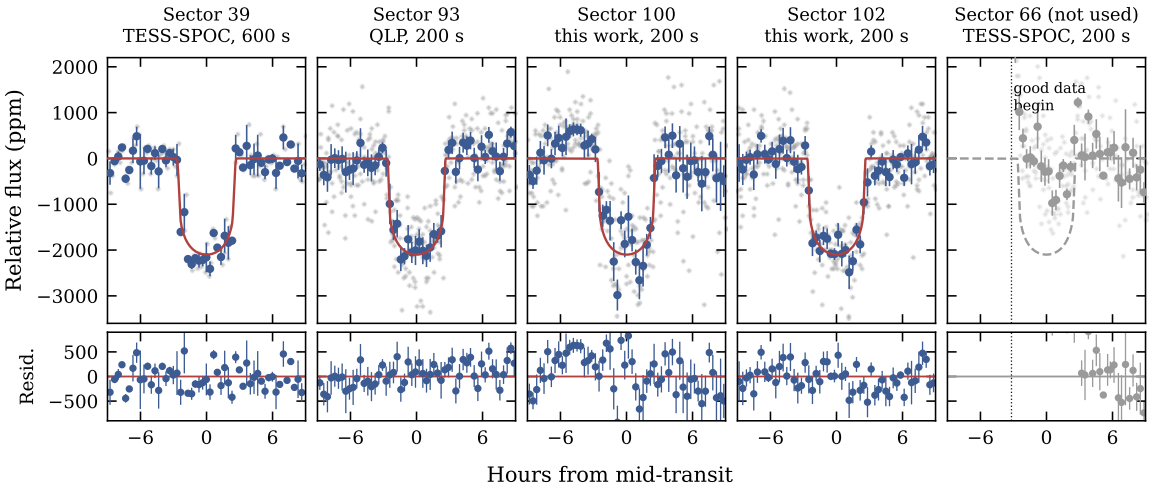


Figure 1: The four transits of TIC 402710280.01 (20-min bins) with the median transit model and residuals. The right-hand panel shows the Sector 66 window, lost to scattered light and excluded from all fits (Vora, 2026a).

References

Burke, C. J., Levine, A., Fausnaugh, M., et al. 2020, `tess-point`, Astrophysics Source Code Library, ascl:2003.001

Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021, AJ, 161, 24

Vora, R. 2026a, preprint, doi:10.5281/zenodo.23390317; data and code, doi:10.5281/zenodo.23390274

Vora, R. 2026b, pre-registered test of the epoch-57 and 59 transits, doi:10.5281/zenodo.23390325